

ShellForge

Linux-Based Administration Console for the Enterprise Knowledge Intelligence Platform

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (OSSP)

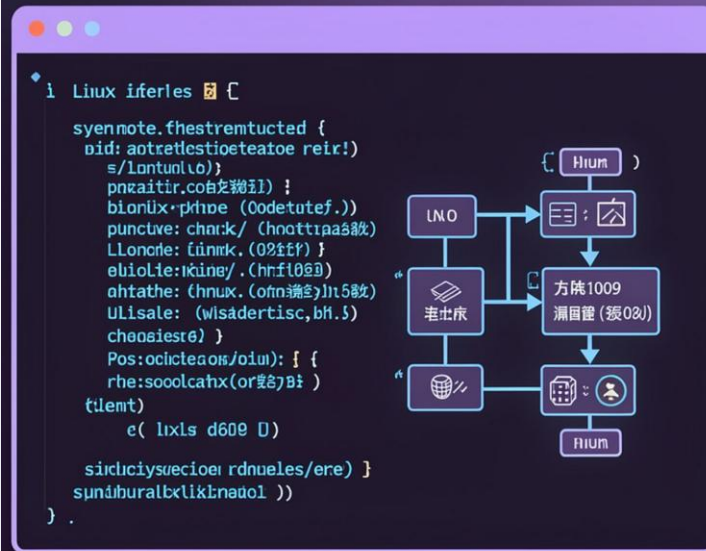
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· Course: OSSP · Guide: Dr. P. Kalpana · KLH University



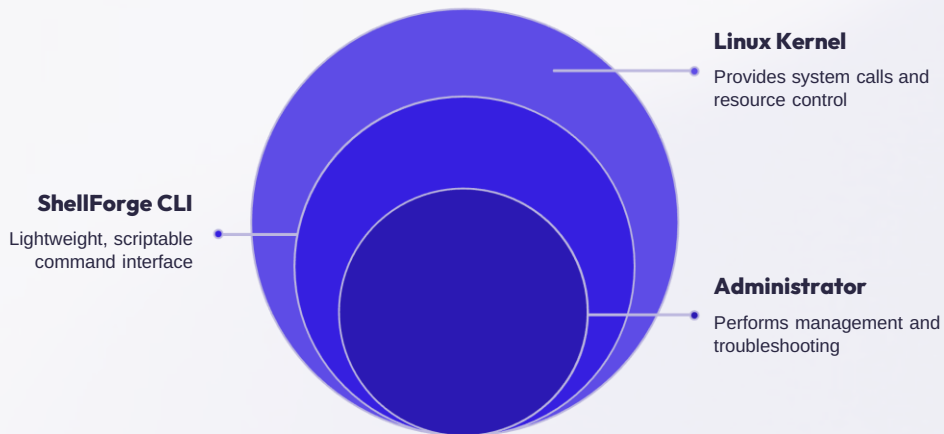
Problem Statement

The Gap

Modern enterprise applications rely on graphical interfaces, but system administrators need a **fast, lightweight, and scriptable CLI** for managing services, monitoring processes, and troubleshooting resources.

The Solution: ShellForge

A **Unix-like shell** built in C that serves as the command-line administration console for the Enterprise Knowledge Intelligence Platform — enabling direct interaction with the backend through Linux system programming concepts.



Project Objectives



Custom Unix Shell

Develop a REPL-based shell in C with command parsing and built-in commands.



Process Management

Implement `fork()`, `exec()`, `wait()`, and background job control.



Memory & File I/O

Manage virtual memory, dynamic allocation, and Linux file descriptors.



IPC & Concurrency

Implement pipes, signals, POSIX threads, mutexes, and synchronization.



Platform Integration

Integrate ShellForge with the Enterprise Knowledge Intelligence Platform.

Technology Stack

Language

C (GNU GCC)

Environment

Ubuntu / WSL · Visual Studio Code · Make · Git & GitHub

Linux APIs

System Calls · pthread · Signals · Pipes · File I/O

Tools

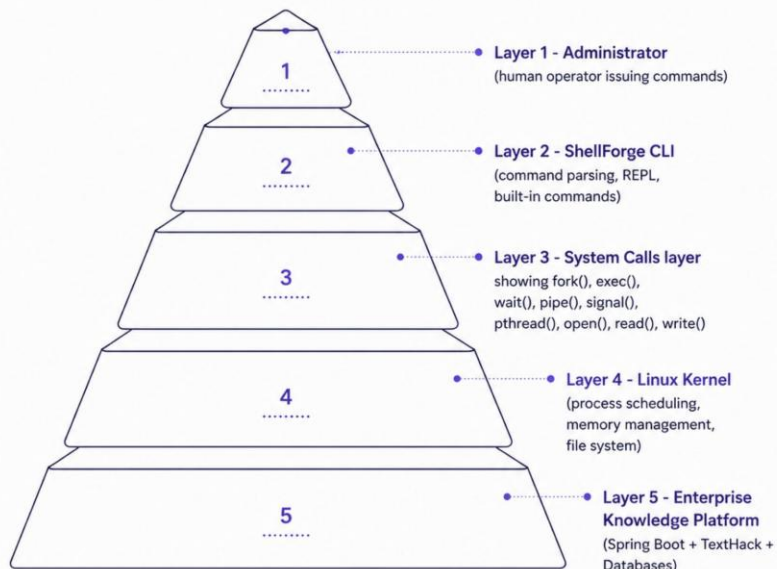
GDB · Valgrind · Makefile · Bash

Development Workflow

ShellForge is developed on **Ubuntu/WSL** using **GNU GCC** and managed via **Git/GitHub**

System Architecture

ShellForge acts as the **Linux administration interface** for the enterprise platform, translating administrator commands into kernel-level system calls.



Layer Breakdown

Administrator

Issues commands via the ShellForge prompt.

ShellForge CLI

Parses input, dispatches system calls.

Linux Kernel

Handles process, memory, and I/O operations.

Enterprise Platform

Spring Boot + TextHack + Databases backend.

Project Modules



Shell & REPL

Command parsing, prompt rendering, built-in commands.



Process Management

fork(), exec(), wait(), background jobs.



IPC

Pipes, named pipes, and signal handling.



Memory Management

Virtual memory, dynamic allocation, debugging.



Linux File System

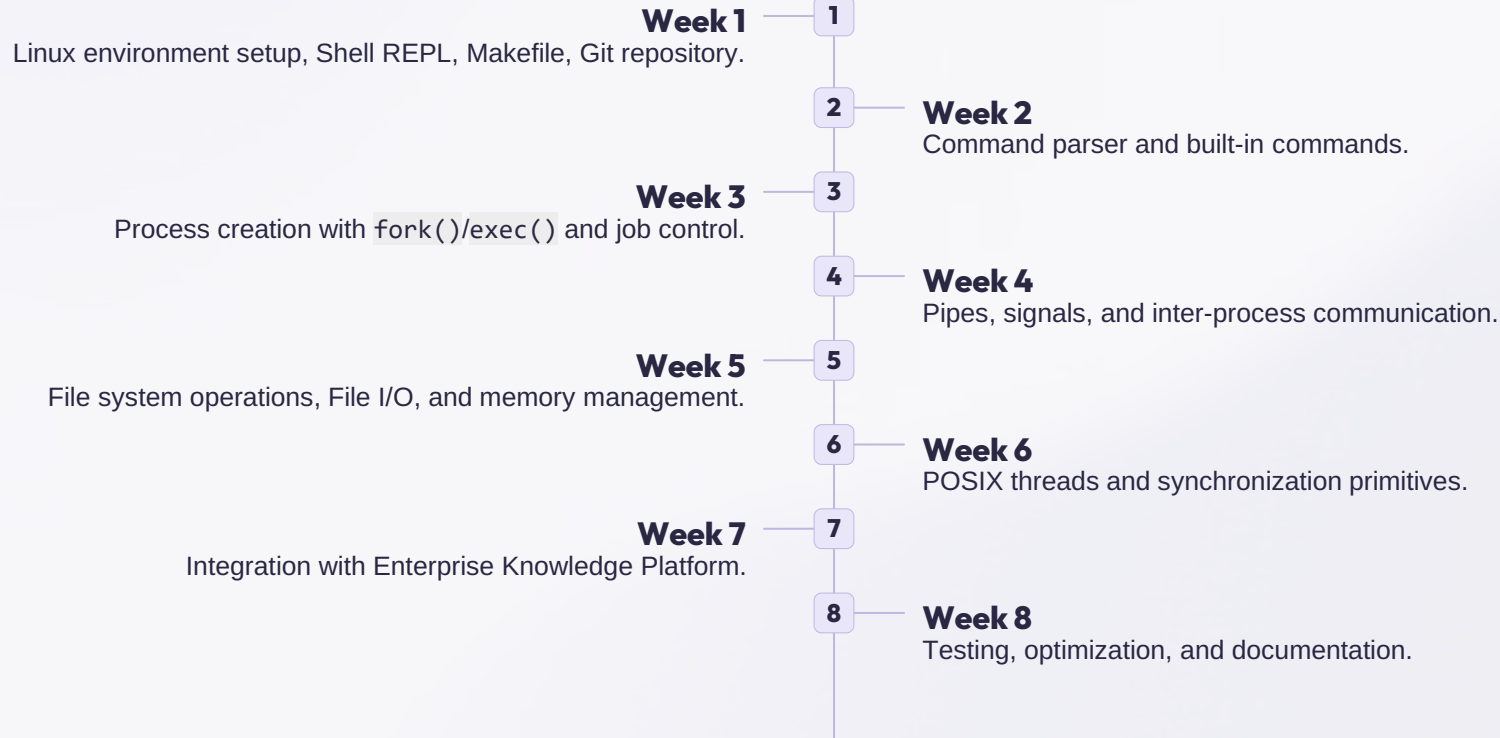
File descriptors, I/O operations, directory management.



Concurrency

POSIX threads, mutex, semaphores, synchronization.

Implementation Timeline



❑ **Expected Outcome:** A fully functional Linux-based administration shell integrated with the Enterprise Knowledge Intelligence Platform, demonstrating practical implementation of process management, IPC, memory management, file systems, and concurrency.